

TIER 1 · CORE PROJECT

LabRack Vision QA

An AI vision QA assistant for lab sample racks

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Why Tier 1: A focused computer-vision application using pretrained YOLO11 and YOLO11-seg with a small staged dataset. Realistic to build in one term, runs on free compute, and can be extended later without overengineering.



The Problem

A visual workflow assistant — not a diagnostic system, and no real patient data.

Labs depend on clean, consistent sample handling. Before samples move through the workflow, staff may need to visually confirm that tubes are present, capped, arranged correctly, and not obviously out of place.

Even small visual issues can create rework, delays, or avoidable quality checks. This project targets lab-rack images only.

Who cares



Lab technicians

Handle and stage samples daily



Accessioning teams

First checkpoint for incoming racks



QA staff

Catch issues before they propagate

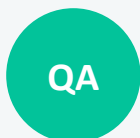


Lab ops managers

Own throughput and rework costs

Your Solution

Analyze a photo of a staged sample rack, detect tubes and rack positions, flag basic visual issues, and generate an annotated QA image plus a short summary.



Example output

“Detected 22 tubes, 1 empty position, 1 possible uncapped tube — confidence review recommended.”

Technical Approach

CV TECHNIQUE

Object detection +
instance segmentation

MODEL

YOLO11 + YOLO11-seg

FRAMEWORK

Ultralytics · Python
OpenCV · Colab / Jupyter

Why this fits

YOLO11 fits because Version 1 needs to find objects in a photo and return visible results quickly. **YOLO11-seg adds masks** so bounding boxes can be compared against cleaner object outlines when the rack or tubes overlap.

Version 1

Detect visible lab objects and produce a useful QA report.

Capstone extension

SAM-style promptable segmentation, video, a dashboard, or richer QA logic.

Data Plan

Self-collected staged photos — empty tubes, racks, fake labels. No PHI.

200–250

images

INITIAL DEMO

100–150

staged images

TRAIN / VALIDATION

50–100

images

HOLDOUT TEST SET

Labels

rack

tube

cap

empty slot

tilted / abnormal tube (if labelable)

Data safety No real patient labels, no real sample IDs, no medical decision-making, no identity claims. Any label text used is fake or blurred.

Success Metrics

PRIMARY

≥ 0.75

mAP50 on the holdout test set for core classes: rack, tube, cap, empty slot.

SECONDARY

$< 3 \text{ sec}$

Per image to produce an annotated output plus QA summary in Colab or local Jupyter.



Practical success check

The demo clearly shows the system detecting rack/tube objects and producing a short, understandable QA summary that a lab worker could review.

Milestone Plan

First working demo comes before heavy data work — prove the pipeline runs before collecting and labeling.

1	Blueprint	Plan approved	Midterm submitted
2	First Working Demo	Run pretrained YOLO11 / YOLO11-seg end-to-end on staged rack images	Something works, even if rough
3	Make It Yours	Add staged rack data, labels, and QA logic (tube count, empty positions, cap checks)	Works on my lab problem
4	Improve & Measure	Test on holdout images, tune confidence thresholds, record mAP / speed	Metrics recorded
5	Package & Present	Clean README, demo notebook, final slides, short demo video	Final submitted

Risks & Resources

Risk 1

Dataset too small or inconsistent

Plan B Narrow V1 to rack, tube, cap — use rule-based counting.

Risk 2

Segmentation masks messy on transparent / reflective tubes

Plan B Use bounding boxes for V1; keep seg as an enhancement.

Risk 3

Real lab images raise privacy / compliance issues

Plan B Staged, non-patient images only — empty tubes, fake labels.

Compute & tools

- Google Colab free GPU or local VSCode / Jupyter
- Ultralytics YOLO11, OpenCV
- Roboflow or Label Studio for labeling
- GitHub for project tracking

Estimated cost: \$0 (\$5–10 optional)

Educational prototype only. Not validated for clinical, diagnostic, or production laboratory use. Human review is required.